USD/JPY Tokenized Settlement Lab

Comparing three settlement designs on synthetic data

Presenter: ssnoa4o · 2026-11

Independent research prototype · synthetic data · not reviewed or endorsed by BIS, BoJ, CLS or any institution

The Problem: Two Different Risks

- An FX trade has two payment legs, for example USD and JPY
- **Principal risk:** one leg settles and the other does not (Herstatt, 1974) [CLS-4]
- **Liquidity risk:** a bank cannot pay on time because a counterparty has not paid yet [OSFI-1]
- PvP is "designed to eliminate principal risk, but will not eliminate a bank's replacement cost risk or liquidity risk" [BCBS-1]
- CLS says its service cuts funding needs by more than 96%. That is CLS's own claim; we did not test it [CLS-3]

Sources: [BCBS-1] Basel Committee 2013; [OSFI-1]; [CLS-3]; [CLS-4]. Full list in the report.

Why Tokenization?

- BIS Project Agorá explores a shared programmable platform with tokenised deposits and tokenised central bank reserves. This is BIS's own description [BIS-2]
- BIS describes a lock, then commit-or-cancel workflow for both legs [BIS-2]
- BIS says its "atomicity" is about the workflow, not legal settlement finality [BIS-2]
- Bank of Japan: tokenized deposits are among the candidates for future foundations (Himino) [BOJ-5]. This is the Bank's view
- Gap: in the parts we read, no USD/JPY PvP test or quantified liquidity result was found

Sources: [BIS-2] Agorá report (read in part); [BOJ-5] Himino speech, 16 May 2026; [BOJ-7] PSS Report 2024.

Three Settlement Models

Model	When funds are checked	When legs move	Can one leg complete alone?
A Sequential	When each leg runs	USD at arrival, JPY after the delay	Yes
B Prefunded	Both legs reserved at arrival	Legs run as in A	Yes, if the second leg fails
C Atomic PvP	Both legs validated at commit	Both applied in one step	No

Prefunding is not atomicity. A balance check alone does not make settlement atomic.

Simplified models. The mapping to real arrangements is our own interpretation.

Research Methodology

Question: Under simplified models and synthetic USD/JPY transactions, how do three designs differ in principal exposure, liquidity demand, time and stress behaviour?

Hypothesis	Claim tested	Metric
H1a, H1b	Exposure: A and B vs failure rate (H1a); B lower than A under JPY stress (H1b)	Principal Exposure
H2	B minus A gap grows with delay	Peak Liquidity Proxy
H3	L* differs by volume, banks, arrival	L*
H4a, H4b	JPY stress changes Success Rate (A vs C); delay stress at JPY 0.09375 changes exposure (A vs B)	Success Rate; Principal Exposure

- Decision rules fixed before the runs; outcomes: supported / not supported / inconclusive
- 30 seeds (1 to 30), same transactions for all models, JPY-only liquidity stress
- Peak Liquidity Proxy is not Peak Liquidity. Gross Prefunding is not Peak Liquidity

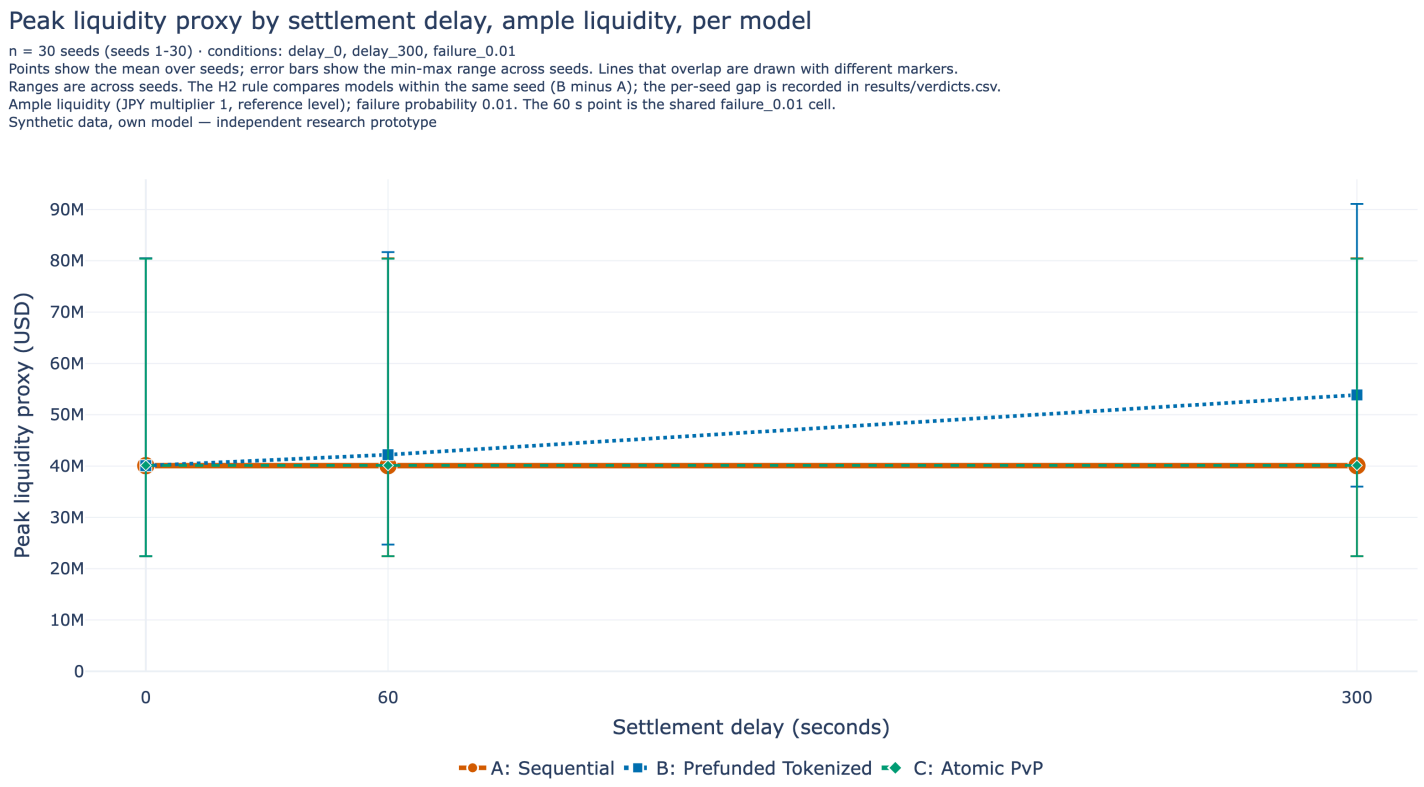
Prototype Demonstration

Streamlit dashboard, runs the real engine locally (no network, no external API)

1. **Baseline:** compare **A Sequential**, **B Prefunded**, **C Atomic PvP**
2. **JPY shortage:** set the multiplier to 0.0625
3. **Look at C:** no one-leg-only state (by design, a correctness check), but it still fails when funds are short
4. **Back to recorded results:** CSV and figures behind the slides

Live runs are exploratory. Verdicts come from the recorded runs (30 seeds), not from the demo. Synthetic data.

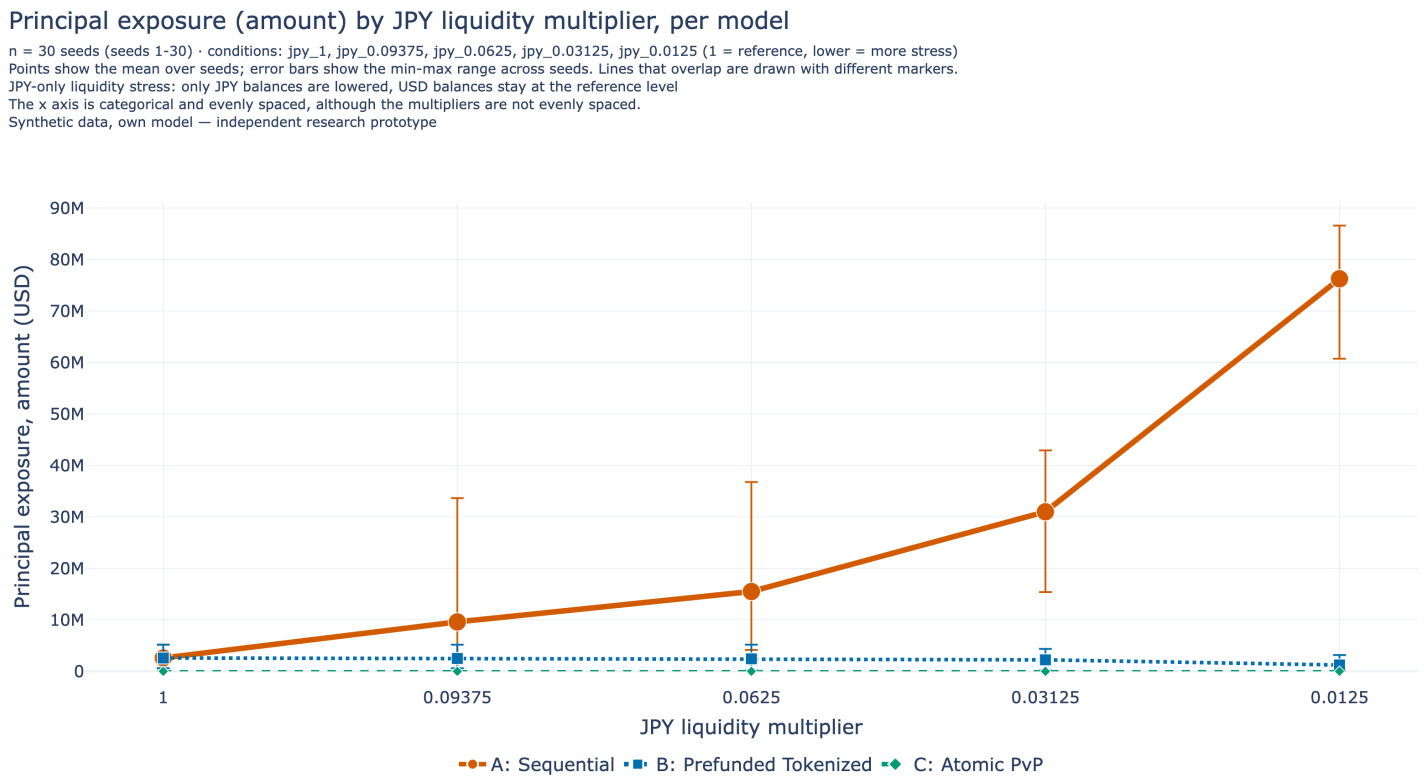
Liquidity Results



Ample liquidity, failure 1%, delays 0, 60, 300 s, 30 seeds, min to max bars. Synthetic data, own model.

- **Result:** B's Peak Liquidity Proxy was 40,070,000 USD at 0 s, 42,210,000 at 60 s, 53,870,000 at 300 s. A stayed at 40,070,000 (4 significant figures)
- **H2 recorded as supported** (gap at 300 s minus gap at 0 s positive in 30 of 30 seeds; mean 13,792,920.307 USD)
- **Caveat:** direction follows from the reservation rule; proxy ignores netting and credit

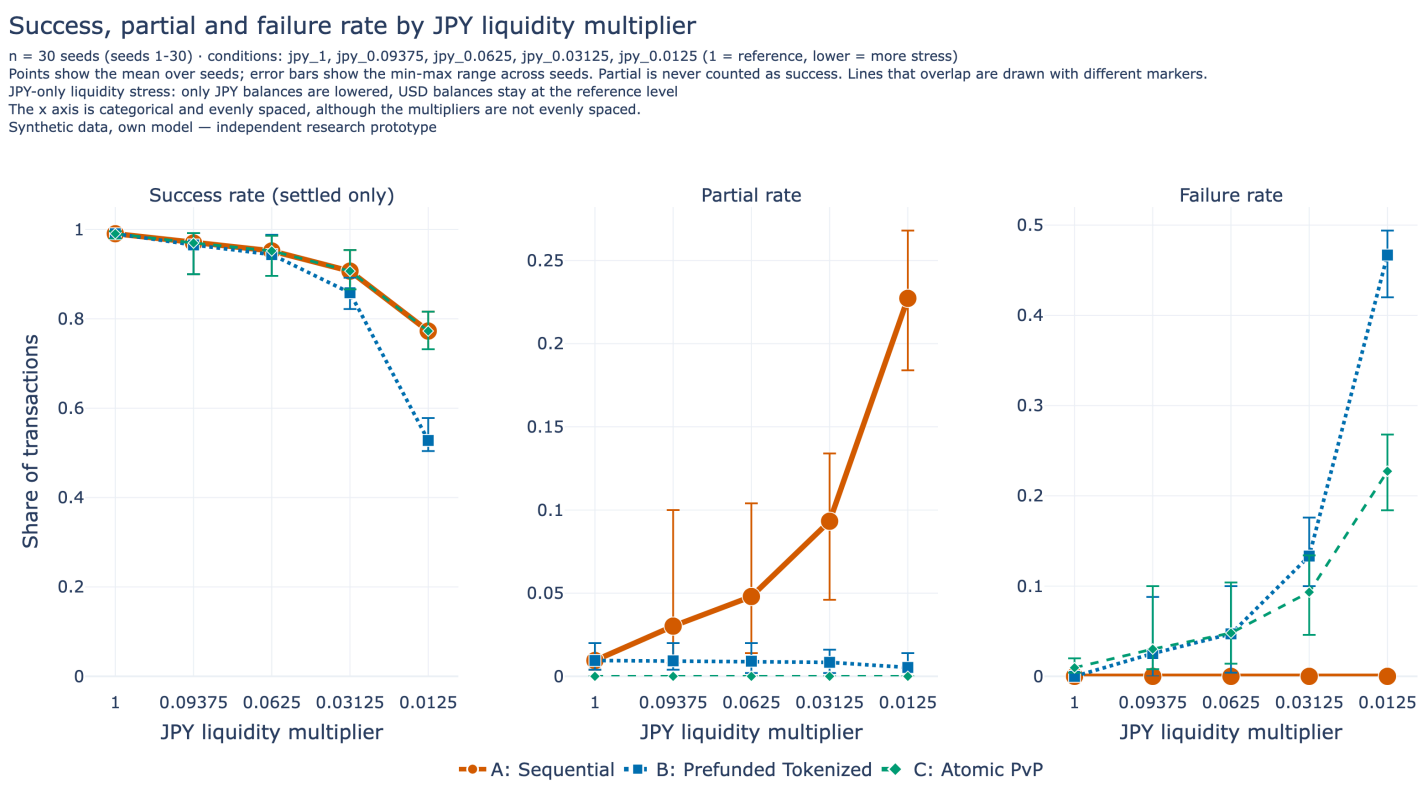
Settlement Risk Results



Amount-based exposure, JPY multipliers 1 to 0.0125, failure 1%, 60 s, 30 seeds, min to max bars. Synthetic data, own model.

- **Baseline:** Success Rate 0.9905 in all models; exposure 2,596,000 USD in A and B, 0 in C
- **H1b recorded as supported:** at 0.0125, A 76,250,000 vs B 1,230,000 USD
- **Caveat:** direction anticipated from the reservation rule. C = 0 holds by design, a correctness check, not evidence

Stress Testing



- **H4a not supported:** A and C had equal Success Rate in all 150 pairs; may follow from model rules
- **H4b inconclusive:** mixed signs at 300 s, reversal at 60 s
- **H3:** 0 supported, 2 not supported, 7 inconclusive (coarse grid)
- C still failed (by design): Failure Rate 0.2273 at multiplier 0.0125

JPY multipliers 1 to 0.0125, failure 1%, 60 s, 30 seeds, min to max bars. Synthetic data, own model.

Policy Implications

What the runs record

- At multiplier 0.0125: Success Rate 0.7727 in A and C; C Failure Rate 0.2273, A Partial Rate 0.2273, B Failure Rate 0.4669 (means, 30 seeds)
- C has no one-leg-only state by design (a correctness check), yet it still failed when JPY was short

Interpretation (tentative, these models only)

- Principal exposure and liquidity demand behave as different quantities

Open policy questions (no recommendation)

- How would settlement asset type interact with reservation and atomic commit?
- Would a platform's liquidity need matter to participants, as [BOJ-7] suggests?
- How would an atomic workflow align with legal finality rules?

Limitations and Future Research

- Simplified models: no netting, credit, queues, retries or price moves
- Synthetic data; one seed set; ranges are not confidence intervals
- Stress lowers JPY only; H3 grid is coarse
- Smart-contract-style completion is not legal settlement finality
- In these models, Atomic PvP removes the one-leg-only state, not liquidity risk (this follows from the model rules)
- **Next tests:** finer grid; USD or two-currency stress; netting and credit; correlated failures

Conclusion

Item	Recorded outcome
H1a	descriptive only
H1b	supported
H2	supported
H3	0 supported, 2 not supported, 7 inconclusive
H4a	not supported
H4b	inconclusive

- **Interpretation, under these models:** principal risk and liquidity need are different quantities
- Verdicts carry caveats: several directions follow from model design
- Code and data: github.com/ssnoa4o/usd-jpy-tokenized-settlement-lab
- **Next:** finer grid, USD stress, netting, legal finality